

Report Period	2026-08-28 - 2026-09-04
Generated At	2026-09-04 13:08:07
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Report Period	2026-08-28 - 2026-09-04	Total Flows Observed	73660
Total Threats Mitigated	29	Total Packets Analyzed	12011566448
High Priority	5	Malicious Packets	12005009283
Low Priority	24	True Negatives Passed	60342
Manual Releases	0		
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	37	ICMP Flood	0
FP Rate	0.06%	SYN Flood	17
IF Precision	99.72%	UDP Flood	5
IF Recall	92.40%	Uncertain	7
IF F1-Score	95.92%		
IF Accuracy	98.47%	Action	Count
		Quarantined	5
		Time Ban	4
		Blackhole	0
		Released	6

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.72%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.40%	Share of actual attacks successfully flagged
F1-Score	95.92%	Balanced measure combining Precision and Recall
Accuracy	98.47%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.06%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.60%	Actual attacks that went undetected
True Positive Rate (TPR)	92.40%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.94%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 13281	FN 1093
Actual: Normal	FP 37	TN 59249

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	13281	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	493.4 ms	Average time taken to detect an attack
Mitigation Response Time	0.2 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.51%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.49%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.26%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	4	4	2026-09-04 13:02:33	2026-09-04 13:05:45
10.0.0.16	3	3	2026-09-04 13:02:26	2026-09-04 13:05:32
10.0.0.18	3	3	2026-09-04 13:02:30	2026-09-04 13:05:36
10.0.0.19	3	3	2026-09-04 13:02:27	2026-09-04 13:05:33
10.0.0.8	2	2	2026-09-04 12:59:48	2026-09-04 13:01:32
10.0.0.6	1	1	2026-09-04 12:59:48	2026-09-04 13:00:49
10.0.0.10	1	1	2026-09-04 12:59:45	2026-09-04 12:59:56
10.0.0.5	1	1	2026-09-04 12:59:45	2026-09-04 12:59:56
10.0.0.7	1	1	2026-09-04 12:59:46	2026-09-04 12:59:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 12:59:48	10.0.0.6	DDoS	UDP Flood	90.6%	High	Time Ban (1m)
2026-09-04 12:59:48	10.0.0.8	DDoS	UDP Flood	90.9%	High	Time Ban (1m)
2026-09-04 12:59:56	10.0.0.5	DDoS	Uncertain	59.6%	Low	Released
2026-09-04 12:59:56	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-04 12:59:56	10.0.0.7	DDoS	Uncertain	68.4%	Low	Released
2026-09-04 13:00:49	10.0.0.6	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:00:49	10.0.0.8	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:01:11	10.0.0.8	DDoS	UDP Flood	90.9%	High	Quarantined
2026-09-04 13:02:26	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:27	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:30	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:32	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	90.2%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-04 13:03:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-04 13:03:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-04 13:03:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-04 13:03:39	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Released
2026-09-04 13:03:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (2m)
2026-09-04 13:05:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (5m)
2026-09-04 13:05:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (5m)
2026-09-04 13:05:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-04 13:05:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (5m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6242	94.6%	Critical	Phase 2	4	2m 5s	2026-09-04 13:05:45	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6142	93.1%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:36	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6325	94.0%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6218	93.3%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:32	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6203	94.6%	Critical	Phase 2	3	5s	2026-09-04 13:03:45	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6175	94.6%	Critical	Phase 2	1	1m 6s	2026-09-04 13:03:39	Ban Expired
10.0.0.18	SYN Flood	0.6317	93.1%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:36	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6402	94.0%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6245	93.3%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:32	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6282	92.7%	Critical	Phase 2	1	6s	2026-09-04 13:02:39	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6422	93.1%	Critical	Phase 2	1	6s	2026-09-04 13:02:36	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6525	90.2%	Critical	Phase 2	1	6s	2026-09-04 13:02:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6360	93.1%	Critical	Phase 2	1	6s	2026-09-04 13:02:32	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5811	90.9%	High	Phase 1	2	21s	2026-09-04 13:01:32	Attack Stopped
10.0.0.6	UDP Flood	0.5836	90.9%	High	Phase 2	1	1m 1s	2026-09-04 13:00:49	Ban Expired
10.0.0.8	UDP Flood	0.5791	90.9%	High	Phase 2	1	1m 1s	2026-09-04 13:00:49	Ban Expired
10.0.0.5	Uncertain	0.6137	59.6%	Low	Phase 1	1	11s	2026-09-04 12:59:56	Attack Stopped
10.0.0.10	Uncertain	0.6182	68.7%	Low	Phase 1	1	11s	2026-09-04 12:59:56	Attack Stopped
10.0.0.7	Uncertain	0.6112	68.4%	Low	Phase 1	1	10s	2026-09-04 12:59:56	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE